

STAT 5200, Notes 3¹

Minchul Shin²

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Readings. Read section 4.1 and 4.2 in W (for now skip Section 4.2.4). These deal with ordinary least squares (OLS) estimation of the single equation linear model. Model assumptions are presented, and the asymptotic consistency and normality of the OLS parameter estimates are discussed. Further, heteroscedasticity is addressed. It is useful to consider how the approach Wooldridge presents differs from that of the classical linear model with fixed covariates, and what is the consequence of such an approach (require large sample size and focus on use of asymptotic theory). As you read Section 4.1 and 4.2, note how Wooldridge uses results he cites in Chapter 3.

The Single Equation Linear Model. Wooldridge views a structural model as one which defines a causal relationship. This is the model we would like to deal with. In many cases a structural model is not estimable as given, and we need to introduce some modifications to arrive at an *estimable model*.

We begin with an uncomplicated setting. We consider the population model to be a linear projection,

$$y = \beta_0 + \beta_1 x_1 + \cdots + \beta_J x_J + u = \mathbf{x}\beta + u, \quad (1)$$

where $y, x_1, \dots, x_J$ are random variables we observe, u is an unobservable random variable defined by (1) (called the disturbance or the error), $\mathbf{x}$ is the row vector $(1, x_1, \dots, x_J)$ consisting of a constant (for the intercept in the model) and the explanatory variables, and β is the column vector $(\beta_0, \beta_1, \dots, \beta_J)'$ of model parameters. The error term can involve, among other things, measurement error or omitted explanatory variables. (Recall that I am using Wooldridge's notation, which makes $\mathbf{x}$ a row vector, and that all other vectors are column vectors.)

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²Email: mincshin@wharton.upenn.edu

We want to estimate β . We assume that a random sample of size N from the population is available. In particular, we assume that $\{(\mathbf{x}_i, y_i) : i = 1, \dots, N\}$ are independent and identically distributed. Here $\mathbf{x}_i = (1, x_{i1}, \dots, x_{iJ})$, the explanatory variables for the i th unit, and we can write

$$y_i = \mathbf{x}_i\beta + u_i, \quad i = 1, \dots, N. \quad (2)$$

By contrast, in the classical linear model with fixed explanatory variables, one assumes that the errors u_i have mean zero (this assumption poses no restriction when the model has an intercept, as here) and are uncorrelated. To perform exact inference (as opposed to asymptotic inference) in the classical linear model, one also assumes the errors are normally distributed. Small sample size is permitted. Typically, the errors are also assumed to have constant variance (homoscedasticity). Thus, in the classical setting, the error term u has an existence of its own and is a driving source in the model.

Assumptions. In our setting of random explanatory variables, we have several choices for assumptions about the model to ensure that OLS consistently estimates β . Recall that an estimator is consistent if it converges in probability to the correct value as N tends to infinity (Definitions 3.3 and 3.8 in W).

We follow the development in W and examine two alternative assumptions.

We collect all regressors in a row vector $\mathbf{x} = [1, x_1, \dots, x_J]$ and denote the length of $\mathbf{x}$ to be K (i.e., $K = 1 + J$).

Assumption 1 (Wooldridge's OLS.1 and OLS.2).

$$(OLS.1) \quad E[\mathbf{x}'u] = \mathbf{0}, \quad (3)$$

$$(OLS.2) \quad rank(E[\mathbf{x}'\mathbf{x}]) = K. \quad (4)$$

Note that $\mathbf{0}$ is a column of zeros of length K .

The first element of $\mathbf{x}$ is 1; therefore (3) implies $E[u] = 0$. Therefore (3) implies that u is uncorrelated with each of the explanatory variables, a condition that is needed. Assumption (4) states that there are no exact linear relations among the explanatory variables (1 included) in the population; that is, there are no redundancies.

Assumption 2. $E[u|\mathbf{x}] = 0$, and (4) also holds.

Assumption 2 is stronger than Assumption 1. Note that $E[u|\mathbf{x}] = 0$ implies (3): By iterated expectations,

$$E[u] = E[E[u|\mathbf{x}]] = E[0] = 0$$

and

$$E[x_j u] = E[E[x_j u|\mathbf{x}]] = E[x_j E[u|\mathbf{x}]] = E[x_j \cdot 0] = 0, \quad j = 1, \dots, J.$$

Note that (1) and (3) imply that $\beta_0 + \beta_1 x_1 + \dots + \beta_J x_J$ is the linear projection of y onto $1, x_1, \dots, x_J$. This follows directly from the definition of the linear projection and (1) and (3).

By contrast, (1) and $E[u|\mathbf{x}] = 0$ imply

$$E[y|1, x_1, \dots, x_J] = \beta_0 + \beta_1 x_1 + \dots + \beta_J x_J,$$

thus that $\beta_0 + \beta_1 x_1 + \dots + \beta_J x_J$ is the conditional expectation of y , given $1, x_1, \dots, x_J$. Note the distinction.

Some terminology: An explanatory variable x_j in (1) is called *exogenous* if it is uncorrelated with u . Thus, if (3) or $E[u|\mathbf{x}] = 0$ holds, all of the explanatory variables are exogenous. An explanatory variable is *endogenous* in (1) if it is correlated with u . There are consequences if explanatory variables are not exogenous, and we will explore these.

Endogeneity. Endogeneity occurs for several reasons, and Wooldridge discusses three common causes at the end of Section 4.1. Often one wants to include an explanatory variable in a regression model, but data for this variable are not available. This is an omitted variable situation. If one fits the model with just the available explanatory variables, the omitted variable becomes part of the error term u , and typically u is then correlated with the explanatory variables included in the model (because the included explanatory variables are usually correlated with the missing explanatory variable). This leads to bias in estimation of the partial slopes attached to *all* the included explanatory variables, regardless of how many correlations there are.

A second cause of endogeneity is measurement error in an explanatory variable. The measurement error is absorbed into the error term u , and this can lead to correlation between u and the imperfectly observed explanatory variable. A third cause of endogeneity is simultaneity, where an explanatory variable is partly determined by the response y (which variable is the response and which is the explanatory variable?).

Let's consider the classic example given by Working (What do 'statistical demand' curves show? *Quarterly Journal of Economics*, Vol. 41 (1927), pp. 212-235). Working considered a simple model

of supply and demand. Let q^d represent the amount demanded for a certain commodity, q^s the amount supplied, and p the price. Write the demand equation as

$$q^d = \alpha_0 + \alpha_1 p + u, \quad (5)$$

and the supply equation as

$$q^s = \beta_0 + \beta_1 p + v, \quad (6)$$

where u and v are error terms which presumably represent variables other than price which affect demand and supply, respectively. Assume the two error terms have mean 0 and are uncorrelated. At market equilibrium, $q^d = q^s = q$, say, and (5) and (6) can be written as

$$q = \alpha_0 + \alpha_1 p + u \quad (5')$$

and

$$q = \beta_0 + \beta_1 p + v. \quad (6')$$

It follows that the explanatory variable p is endogenous in both (5) and (6). The solution for p from (5) and (6) is

$$p = \frac{\beta_0 - \alpha_0}{\alpha_1 - \beta_1} + \frac{v - u}{\alpha_1 - \beta_1}, \quad (7)$$

a function of both error terms. Thus

$$Cov(p, u) = -\frac{Var(u)}{\alpha_1 - \beta_1}, \quad Cov(p, v) = \frac{Var(v)}{\alpha_1 - \beta_1}. \quad (8)$$

The endogeneity is a result of market equilibrium.

OLS Estimation and Consistency of the Estimator. Consider estimation of the parameters in (2), under Assumption 1 and a given random sample $\{(y_i, \mathbf{x}_i) : i = 1, \dots, N\}$. Recall that $\mathbf{x}_i = (1, x_{i2}, \dots, x_{iK})$.

First let's show that the parameter vector β is identified, that is, that it can be expressed in terms of population moments of observed variables. Pre-multiply $y = \mathbf{x}\beta + u$ by $\mathbf{x}'$ to obtain

$$\mathbf{x}'\mathbf{x}\beta + \mathbf{x}'u = \mathbf{x}'y.$$

Taking the expectation, we obtain

$$E[\mathbf{x}'\mathbf{x}]\beta = E[\mathbf{x}'y], \quad \text{or} \quad \beta = [E[\mathbf{x}'\mathbf{x}]]^{-1} E[\mathbf{x}'y]. \quad (9)$$

The second equation in (9) gives the parameter vector.

Using the method of moments, we replace the two expectations in (9) by their sample averages, obtaining the estimator

$$\hat{\beta} = \left(\frac{1}{N} \sum_{i=1}^N \mathbf{x}'_i \mathbf{x}_i \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \mathbf{x}'_i y_i \right). \quad (10)$$

By (4) of Assumption 1, the inverse matrix in (10) exists with probability tending to 1 as N tends to infinity. Specifically, by the law of large numbers (Theorem 3.1 in W) and Corollary 3.1 in W, the probability limit of the first term is

$$\left(\frac{1}{N} \sum_{i=1}^N \mathbf{x}'_i \mathbf{x}_i \right)^{-1} \rightarrow_p [E[\mathbf{x}' \mathbf{x}]]^{-1} = \mathbf{A}^{-1},$$

where we are using the notation $\mathbf{A} = E[\mathbf{x}' \mathbf{x}]$.

By (2), expression (10) can be written as

$$\hat{\beta} = \beta + \left(\frac{1}{N} \sum_{i=1}^N \mathbf{x}'_i \mathbf{x}_i \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \mathbf{x}'_i \mathbf{u}_i \right). \quad (11)$$

By the law of large numbers,

$$\frac{1}{N} \sum_{i=1}^N \mathbf{x}'_i \mathbf{u}_i \rightarrow_p E[\mathbf{x}' \mathbf{u}] = 0,$$

and therefore, by Lemma 3.4 and Corollary 3.1 in W,

$$\hat{\beta} \rightarrow_p \beta$$

under the given assumptions. That is, OLS estimation provides consistent estimation of the parameters of a linear projection under the given assumptions.

Note that the estimator $\hat{\beta}$ also arises from minimizing

$$\sum_{i=1}^N (y_i - \mathbf{x}_i \beta)^2$$

that is, it is the OLS estimator.

Next let's observe how to write (10) and (11) in matrix-vector form, as is done for the classical linear model with fixed covariates. Define

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_N \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_N \end{pmatrix},$$

and $N \times K$ matrix and two $N \times 1$ vectors. Then (10) and (11) are

$$\hat{\beta} = (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \mathbf{y} = \beta + (\mathbf{X}' \mathbf{X})^{-1} \mathbf{X}' \mathbf{u}.$$

Asymptotic Normality of the OLS Estimator and Inference. From the above we can write for the OLS estimator

$$\sqrt{N}(\hat{\beta} - \beta) = \left(\frac{1}{N} \sum_{i=1}^N \mathbf{x}'_i \mathbf{x}_i \right)^{-1} \left(\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{x}'_i u_i \right). \quad (12)$$

As N tends to infinity, the first factor on the right-hand side converges in probability to $\mathbf{A}^{-1}$, where $\mathbf{A} = E[\mathbf{x}'\mathbf{x}]$, and the second factor converges in distribution, by the central limit theorem, to $N(\mathbf{0}, \mathbf{B})$, where $\mathbf{B} = E[u^2 \mathbf{x}'\mathbf{x}]$ (e.g., Theorem 3.2 in W). Thus (12) converges in distribution to (use Slutsky's theorem)

$$N(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1})$$

For more details about this, see Appendix 1 to these notes.

The next assumption adds a homoscedasticity condition.

Assumption 3. $E[u^2 \mathbf{x}'\mathbf{x}] = \sigma^2 E[\mathbf{x}'\mathbf{x}]$, where $\sigma^2 = E[u^2] = Var(u)$.

Under this further assumption, $\mathbf{B}$ is equal to $\sigma^2 \mathbf{A}$, and thus (12) converges in distribution to $N(\mathbf{0}, \sigma^2 \mathbf{A}^{-1})$. With this assumption, it is customary to use the classical model OLS standard errors, t statistics and F statistics. We estimate the error variance as

$$\hat{\sigma}^2 = \frac{1}{N - K} \sum_{i=1}^N \hat{u}_i^2,$$

where $\hat{u}_i$ is the i th residual. If the divisor $N - K$ is replaced by N , one has the maximum likelihood estimator of the error variance, and for large samples either divisor can be used. The matrix $\mathbf{A}$ is estimated by $\mathbf{X}'\mathbf{X}/N$, and we treat $\hat{\beta}$ as

$$N(\beta, \hat{\mathbf{V}}),$$

where

$$\widehat{Avar}(\hat{\beta}) = \hat{\sigma}^2 (\mathbf{X}'\mathbf{X})^{-1} \equiv \hat{\mathbf{V}} = (\hat{v}_{jk}).$$

It is common to focus attention on one of the components of $\hat{\beta}$, say $\hat{\beta}_k$. We treat $\hat{\beta}_k$ as $N(\beta_k, \hat{v}_{kk})$. The t -statistic for testing the null hypothesis $H_0 : \beta_k = \beta_{k0}$ is

$$\frac{\hat{\beta}_k - \beta_{k0}}{\sqrt{\hat{v}_{kk}}},$$

and a $100(1 - \alpha)$ per cent confidence interval for β_k is $\hat{\beta}_k \pm z_{\alpha/2} \sqrt{\hat{v}_{kk}}$, where $z_{\alpha/2}$ is the $100(1 - \alpha/2)$ quantile of the standard normal distribution. This use of the normal quantile instead of the quantile

of the t -distribution with $N - K$ degrees of freedom assumes that N is large. In fact, R always uses the t -distribution quantile.

One should note that use of t and F statistics here is, in a sense, adoption of the classical linear model assumptions, rather than sticking to Assumption 1 or Assumption 2 and asymptotic theory. If we stick to asymptotic theory, tests of parameter values are chi-square tests (Wald statistics), and confidence intervals and regions are constructed with normal distribution and chi-square distribution quantiles, rather than t -distribution and F -distribution quantiles. For more details, see Appendix 2 in these notes.

If Assumption 3 does not hold, the standard errors are obtained from estimation of sandwich expression $\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$. Details are in Appendix 2 of these notes, and will also be discussed in the next set of notes. Some comments on residuals are in Appendix 3.

Should Heteroscedasticity Be Expected? Weighted Least Squares. Suppose the correct model is given by a conditional expectation. That is, $y = E[y|\mathbf{x}] + u$. Further, suppose that one uses the linear projection (linear regression) $\mathbf{x}\beta$ to approximate the conditional expectation. Then

$$y - \mathbf{x}\beta = y - E[y|\mathbf{x}] + E[y|\mathbf{x}] - \mathbf{x}\beta = u + E[y|\mathbf{x}] - \mathbf{x}\beta,$$

and

$$E[(y - \mathbf{x}\beta)^2|\mathbf{x}] = \text{Var}(y|\mathbf{x}) + 2E[u(E[y|\mathbf{x}] - \mathbf{x}\beta)|\mathbf{x}] + E[(E[y|\mathbf{x}] - \mathbf{x}\beta)^2|\mathbf{x}].$$

This states that the residual variance in a linear regression tends to increase as the difference between the true model (the conditional expectation) and the fitted model (the linear projection) increases. That is, homoscedasticity is unlikely – we should expect the presence of heteroscedasticity.

If Assumption 3 does not hold, we have heteroscedasticity. That is, if $E[u^2|\mathbf{x}]$ does not equal $E[u^2] = \sigma^2$, $\mathbf{B} = E[u^2\mathbf{x}'\mathbf{x}]$ does not reduce to $\sigma^2E[\mathbf{x}'\mathbf{x}]$. The consequence is that we cannot use the standard errors commonly provided by statistical programs. In many situations this is not a problem – one simply uses standard errors obtained from the sandwich expression $\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$, given below (12). There are situations, however, where one wants to make an adjustment so that homoscedasticity is induced. For example, in some cases one can transform the response and explanatory variables to achieve this end.

Weighted least squares is a methodology that is widely used, often with the aim of inducing homoscedasticity. But there are other purposes for its use. Consider the model $y = \mathbf{x}\beta + u$, and suppose that

$$E[y|\mathbf{x}] = \mathbf{x}\beta \quad \text{and} \quad E[u^2|\mathbf{x}] = \text{Var}(y|\mathbf{x}).$$

Then,

$$y^* = \mathbf{x}^* \beta + u^*,$$

where

$$y^* = y / \sqrt{\text{Var}(y|\mathbf{x})}, \quad \mathbf{x}^* = \mathbf{x} / \sqrt{\text{Var}(y|\mathbf{x})}, \quad u^* = u / \sqrt{\text{Var}(y|\mathbf{x})},$$

is a revision of the model with homoscedasticity incorporated, and the least squares solution for the parameter vector in this formulation is obtained from minimization with respect to β of the expression

$$E[(y^* - \mathbf{x}^* \beta)^2] = E[(y - \mathbf{x} \beta)^2 / \text{Var}(y|\mathbf{x})].$$

The result is

$$\beta_{WLS} = [E[(\mathbf{x}^{*'} \mathbf{x}^*)]]^{-1} E[\mathbf{x}^{*'} y^*],$$

with the subscript attached to emphasize that the weighted least squares solution is usually different from the OLS solution

$$\beta_{OLS} = [E[\mathbf{x}' \mathbf{x}]]^{-1} E[\mathbf{x}' y].$$

With a random sample of data $\{(y_i, \mathbf{x}_i) : i = 1, \dots, N\}$, one performs weighted least squares by minimizing with respect to β the expression

$$\sum_{i=1}^N (y_i - \mathbf{x}_i \beta)^2 w(\mathbf{x}_i),$$

where $w(\mathbf{x}_i) = 1/\text{Var}(y|\mathbf{x}_i)$. Of course, this requires knowledge of the conditional variance of y for each value $\mathbf{x}_i$, and such knowledge is usually not available. If there are multiple values of y for each value of $\mathbf{x}$, one can estimate the values $\text{Var}(y|\mathbf{x})$. However, such estimates may not be very reliable, especially if there are only a few values of y for some values of $\mathbf{x}$. If this approach is reliable, though, one can use the usual standard errors stemming from a homoscedasticity assumption.

Sometimes the values of $\text{Var}(y|\mathbf{x})$ are known up to a scalar constant. For example, suppose that, instead of microdata, you have y averages grouped at the level of the $\mathbf{x}$ values. This typically occurs when the original data violate confidentiality of subjects. Suppose that $\bar{y}(\mathbf{x})$ denotes the group average of the $N(\mathbf{x})$ values of y for which the explanatory variable vector is equal to $\mathbf{x}$, and suppose the $N(\mathbf{x})$ values are known. If the microdata are homoscedastic with conditional variance $\sigma^2 = E[u^2|\mathbf{x}]$, the correct choice of weight is

$$w(\mathbf{x}_i) = N(\mathbf{x}_i), \quad i = 1, 2, \dots, N.$$

In this case, the weighted least squares estimate of the parameter vector using the group averages is identical to the OLS estimate of the parameter vectors using the original microdata. This is the analog in the sample of the result in Problem 2.9 in W. With this estimation, one does not obtain the standard errors corresponding to the OLS estimation with the original microdata, however.

Regression to the Mean. The term regression originated with Sir Francis Galton (1822–1911). Galton studied the heights of parents and their adult children. He observed that on average the children of tall parents have heights less extreme than those of their parents, and that the same is true on average for children of short parents. In other words, there is observed in the next generation a regression toward the mean population height.

Karl Pearson (1857–1936) conducted a statistical study of the heights of 1,078 fathers and their adult sons. Pearson plotted the fathers' heights on the x -axis and the sons' heights on the y -axis. The slope of the least squares straight line for Pearson's data was approximately 0.5, illustrating the regression toward the mean phenomenon. Note that sons tended to be comparable to those of their fathers, relative to the mean of the population within each generation.

The regression to the mean phenomenon typically occurs in many applications. One is a course with successive examinations. Those who score high (low) relative to the middle on the first examination tend to score less high (less low) relative to the middle on the second examination.

Why does regression to the mean occur in certain contexts? The slope in the linear projection for the model $y = \beta_0 + \beta_1 x + u$ is

$$\beta_1 = \frac{\text{Cov}(x, y)}{\text{Var}(x)}$$

In the context of fathers and their sons, if y is the son's height and x is the father's height, $\text{Var}(x)$ and $\text{Var}(y)$ are approximately equal. Then the slope in the linear projection is approximately equal to

$$\frac{\text{Cov}(y, x)}{\sqrt{\text{Var}(y)\text{Var}(x)}},$$

the correlation between the two variables, and thus it is can be expected to be less than 1 (and it is certainly greater than 0). If (y, x) is bivariate normal (a good model for the heights of parents and their children), the conditional expectation $E[y|x]$ is given by

$$E[y|x] = \mu_y + \rho \frac{\sigma_y}{\sigma_x} (x - \mu_x)$$

[see Notes 1]. If we assume $\sigma_y = \sigma_x$, the conditional mean becomes

$$E[y|x] = \mu_y - \mu_x \rho + \rho x,$$

and, if $\mu_y = \mu_x$, this becomes $\mu_y(1 - \rho) + \rho x$.

Wooldridge discusses regression to the mean in Problem 4.9.

Example 1. Problem 4.1 in W. Consider the model for men's wages

$$\log(wage) = \beta_0 + \beta_1 married + \beta_2 educ + \mathbf{z}\gamma + u,$$

where

$$E[u|married, educ, \mathbf{z}] = 0.$$

Thus, all the explanatory variables are exogenous. The variable *married* is categorical (with values 1 and 0), and $\mathbf{z}$ represents factors other than marital status and education.

(a) Let θ_1 be the percentage difference in expected wage between married and unmarried men, holding all other explanatory variables fixed. We have

$$E[\log wage|married = 1, educ, \mathbf{z}] = \beta_0 + \beta_1 + \beta_2 educ + \mathbf{z}\gamma \quad (13)$$

$$E[\log wage|married = 0, educ, \mathbf{z}] = \beta_0 + \beta_2 educ + \mathbf{z}\gamma. \quad (14)$$

The difference, (13) minus (14), is

$$E[\log wage|married = 1, educ, \mathbf{z}] - E[\log wage|married = 0, educ, \mathbf{z}] = \beta_1. \quad (15)$$

The definition of θ_1 is

$$\frac{E[wage|married = 1, educ, \mathbf{z}]}{E[wage|married = 0, educ, \mathbf{z}]} = 1 + \frac{\theta_1}{100}.$$

We cannot use (15) to get a direct handle on the parameter θ_1 , because we cannot interchange the expectation and logarithm operations in (15). If we could make the interchange, (15) would become

$$\begin{aligned} E[\log wage|married = 1, educ, \mathbf{z}] - E[\log wage|married = 0, educ, \mathbf{z}] &= \log \left(\frac{E[wage|married = 1, educ, \mathbf{z}]}{E[wage|married = 0, educ, \mathbf{z}]} \right) \\ &= \log \left(1 + \frac{\theta_1}{100} \right) \\ &= \beta_1, \end{aligned} \quad (16)$$

and then we could derive the result given by Wooldridge. Let's do this – I'll return to the question of the interchange below. Approximate the logarithm by using the first term in its Taylor expansion, which is $\theta_1/100$. Therefore, if β_1 is small, $\theta_1 \approx 100\beta_1$. The exact value is $\theta_1 = 100[\exp(\beta_1) - 1]$.

Now, what about the interchange? The logarithm function is concave, and, given a random variable y , we can use the inequality

$$E[\log(y|\mathbf{x})] \leq \log(E[y|\mathbf{x}])$$

(This follows from the modification of Property CE.7 in W to treat concave functions.) However, this doesn't appear to help us manipulate (15) to obtain (16) or get close to (16), or even derive a useful inequality for (15).

Let's make the independence assumption, as in Notes 1. Let y denote the random variable *wage*, so that

$$\log(y) = \beta_0 + \beta_1 \text{married} + \beta_2 \text{educ} + \mathbf{z}\gamma + u,$$

and assume that the error u is *independent of the explanatory variables*. In his statement of the problem, Wooldridge makes this assumption. With it we have

$$\begin{aligned} y &= \exp(\beta_0 + \beta_1 \text{married} + \beta_2 \text{educ} + \mathbf{z}\gamma + u) \\ &= \exp(\beta_0 + \beta_1 \text{married} + \beta_2 \text{educ} + \mathbf{z}\gamma) \exp(u), \end{aligned}$$

and, conditional on the explanatory variables,

$$E[y | \text{married}, \text{educ}, \mathbf{z}] = \exp(\beta_0 + \beta_1 \text{married} + \beta_2 \text{educ} + \mathbf{z}\gamma) E[\exp(u)]. \quad (17)$$

Using (17), we obtain

$$\begin{aligned} 1 + \frac{\theta_1}{100} &= \frac{E[y | \text{married} = 1, \text{educ}, \mathbf{z}]}{E[y | \text{married} = 0, \text{educ}, \mathbf{z}]} \\ &= \frac{\exp(\beta_0 + \beta_1 + \beta_2 \text{educ} + \mathbf{z}\gamma) E[\exp(u)]}{\exp(\beta_0 + \beta_2 \text{educ} + \mathbf{z}\gamma) E[\exp(u)]} \\ &= \exp(\beta_1). \end{aligned}$$

Interestingly, the interchange of the expectation and logarithm operations in (15) produces the correct answer, under the assumption that the error term in the model is distributed independently of the explanatory variables. For the remainder of this problem we continue with this assumption.

(b) A consistent estimator of θ_1 is $\hat{\theta}_1 = 100[\exp(\hat{\beta}_1) - 1]$, where $\hat{\beta}_1$ is the OLS estimator of β_1 . What is the asymptotic standard error of $\hat{\theta}_1$? The derivative of θ_1 with respect to β_1 is $100 \exp(\beta_1)$. Thus, by the delta method, the asymptotic standard error is

$$se(\hat{\theta}_1) = 100 \exp(\hat{\beta}_1) se(\hat{\beta}_1).$$

(c) Repeat (a) and (b) for θ_2 , the percentage change in expected wage when *educ* changes by the amount Δeduc . We find that $\theta_2 \approx 100\beta_2 \Delta \text{educ}$, an approximation which is valid for small percentage changes. The exact value is $\theta_2 = 100[\exp(\beta_2 \Delta \text{educ}) - 1]$, which we estimate consistently by $\hat{\theta}_2 = 100[\exp(\hat{\beta}_2 \Delta \text{educ}) - 1]$. The asymptotic standard error is

$$100 \Delta \text{educ} \exp(\hat{\beta}_2 \Delta \text{educ}) se(\hat{\beta}_2).$$

(d) Wooldridge asks the reader to repeat (a) and (b) for a model fit to the data set `nls80`, which contains data on men in the labor force, apparently sampled during the year 1980. See separate notes for lab sessions.

Example 2. A model often used to study wages is

$$\log(wage) = \beta_0 + \beta_1 educ + \beta_2 exper + \beta_3 exper^2 + u,$$

where *educ* is the number of years of schooling and *exper* is defined to be age minus years of schooling minus six years, a common assumption in studies of wages. Theory suggests that the earnings function should be concave in *exper*, and that thus the estimates of β_2 and β_3 should be positive and negative, respectively. If we differentiate the log of *wage* with respect to *exper*, we find that the level of *exper* for which $\log(wage)$ is maximized is $exper = -\beta_2/(2\beta_3)$, independent of *educ*. Note that the age at which the *wage* will peak varies with *educ*, given the definition of *exper*.

See separate notes for lab session.

Example 3. In the spirit of the first part of Example 1 (Problem 4.1 (a) in W), let's consider the model

$$\log(y) = \beta_0 + \beta_1 \log(x_1) + \beta_2 x_2 + u, \quad \text{where} \quad E[u|x_1, x_2] = 0.$$

Moreover, assume u is independent of all the explanatory variables. Suppose x_1 changes from x_{10} to $x_{10} + \delta$. What is the percentage change in $E[y|x_1, x_2]$, holding x_2 fixed? Call this percentage θ . Then we write

$$y = \exp(\beta_0 + \beta_1 \log(x_1) + \beta_2 x_2) \exp(u),$$

and thus

$$\begin{aligned} 1 + \frac{\theta}{100} &= \frac{E[y|x_{10} + \delta, x_2]}{E[y|x_{10}, x_2]} \\ &= \frac{\exp(\beta_0 + \beta_1 \log(x_{10} + \delta) + \beta_2 x_2) E[\exp(u)]}{\exp(\beta_0 + \beta_1 \log(x_{10}) + \beta_2 x_2) E[\exp(u)]} \\ &= \exp \left[\beta_1 \log \left(1 + \frac{\delta}{x_{10}} \right) \right] \\ &= (1 + \delta/x_{10})^{\beta_1}. \end{aligned}$$

Therefore,

$$\theta = 100 \left[(1 + \delta/x_{10})^{\beta_1} - 1 \right]. \tag{18}$$

This is the exact expression for the percentage θ . To obtain an approximate expression which is valid for small percentage changes, we use Newton's generalized binomial theorem: For any real number a (in fact, a can be any complex number for Newton's result),

$$(x + y)^a = \sum_{j=0}^{\infty} \binom{a}{j} x^{a-j} y^j, \quad (19)$$

where $\binom{a}{0} = 1$ and

$$\binom{a}{j} = \frac{a(a-1) \cdots (a-j+1)}{j!}. \quad (20)$$

Then an approximate expression for (18) is, for small δ/x_{10} , taking the first two terms in the expansion (19),

$$\theta \approx 100\beta_1\delta/x_{10} = \beta_1 \text{ times the percentage change in } x_1. \quad (21)$$

Note that (18) and (21) are the same results as those cited in Notes 1 for the case labeled (d).

Property LP.7 in W. Property LP.7 in W is a partitioned linear projection result. It is stated in the context of the population. It also holds in the sample setting, of course. Let's consider the sample context, and use matrix-vector notation to prove the result. This will allow us to look at the structure of a specific projection matrix (the sample analog of a population projection operator).

Recall the notation in the earlier part of this document. The model is $y = \mathbf{x}\beta + u$, and we have a random sample $\{(\mathbf{x}_i, y_i) : i = 1, \dots, N\}$ of data. We define

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_N \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} y_1 \\ \vdots \\ y_N \end{pmatrix}, \quad \mathbf{u} = \begin{pmatrix} u_1 \\ \vdots \\ u_N \end{pmatrix},$$

an $N \times K$ matrix and two $N \times 1$ vectors. Then the OLS estimate of β is

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}.$$

I'm going to use different notation from that employed by Wooldridge in describing LP.7. Write the partitioned model as

$$y = \mathbf{x}\beta + u = \mathbf{x}_1\beta_1 + \mathbf{x}_2\beta_2 + u.$$

The corresponding partitions of $\mathbf{X}$ and β are

$$\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2), \quad \beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}.$$

The least squares estimate of β is

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{y}) = \begin{pmatrix} \hat{\beta}_1 \\ \hat{\beta}_2 \end{pmatrix} = \begin{pmatrix} \mathbf{X}'_1\mathbf{X}_1 & \mathbf{X}'_1\mathbf{X}_2 \\ \mathbf{X}'_2\mathbf{X}_1 & \mathbf{X}'_2\mathbf{X}_2 \end{pmatrix}^{-1} \begin{pmatrix} \mathbf{X}'_1\mathbf{y} \\ \mathbf{X}'_2\mathbf{y} \end{pmatrix}. \quad (22)$$

We need an expression for the inverse of the partitioned matrix in (22). Let $\mathbf{A}$ be the partitioned matrix

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix},$$

and denote its inverse by (assume all indicated inverses exist),

$$\mathbf{A}^{-1} = \mathbf{B} = \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{pmatrix}.$$

Then

$$\mathbf{I} = \mathbf{AB} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11}\mathbf{B}_{11} + \mathbf{A}_{12}\mathbf{B}_{21} & \mathbf{A}_{11}\mathbf{B}_{12} + \mathbf{A}_{12}\mathbf{B}_{22} \\ \mathbf{A}_{21}\mathbf{B}_{11} + \mathbf{A}_{22}\mathbf{B}_{21} & \mathbf{A}_{21}\mathbf{B}_{12} + \mathbf{A}_{22}\mathbf{B}_{22} \end{pmatrix}.$$

It follows that

$$\mathbf{A}_{22}^{-1}\mathbf{A}_{21} = -\mathbf{B}_{21}\mathbf{B}_{11}^{-1} \quad \text{and} \quad \mathbf{A}_{11}^{-1}\mathbf{A}_{12} = -\mathbf{B}_{12}\mathbf{B}_{22}^{-1}. \quad (23)$$

Multiply $\mathbf{I} = \mathbf{AB}$ on the right by $\mathbf{B}_{11}^{-1}$ to obtain

$$\mathbf{B}_{11}^{-1} = \mathbf{A}_{11} + \mathbf{A}_{12}\mathbf{B}_{21}\mathbf{B}_{11}^{-1} = \mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21}, \quad (24)$$

and write

$$\mathbf{G}_1 = (\mathbf{A}_{11} - \mathbf{A}_{12}\mathbf{A}_{22}^{-1}\mathbf{A}_{21})^{-1}. \quad (25)$$

Now assume that $\mathbf{A}$ is symmetric, so that $\mathbf{A}'_{21} = \mathbf{A}_{12}$ and $\mathbf{B}'_{21} = \mathbf{B}_{12}$. Transpose the first equation in (23) to obtain

$$\mathbf{A}_{12}\mathbf{A}_{22}^{-1} = -\mathbf{B}_{11}^{-1}\mathbf{B}_{12}. \quad (26)$$

(the diagonal submatrices are all symmetric). It follows that

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{pmatrix} = \begin{pmatrix} \mathbf{G}_1 & -\mathbf{G}_1\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \\ -\mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{G}_1 & \mathbf{A}_{22}^{-1} + \mathbf{A}_{22}^{-1}\mathbf{A}_{21}\mathbf{G}_1\mathbf{A}_{12}\mathbf{A}_{22}^{-1} \end{pmatrix}. \quad (27)$$

Note that, alternatively, $\mathbf{B}_{22} = (\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12})^{-1}$.

Now let $\mathbf{A}$ be $\mathbf{X}'\mathbf{X}$, and $\mathbf{B}$ be $(\mathbf{X}'\mathbf{X})^{-1}$, so that

$$\mathbf{A} = \begin{pmatrix} \mathbf{X}'_1\mathbf{X}_1 & \mathbf{X}'_1\mathbf{X}_2 \\ \mathbf{X}'_2\mathbf{X}_1 & \mathbf{X}'_2\mathbf{X}_2 \end{pmatrix}. \quad (28)$$

Here $\mathbf{X} = (\mathbf{X}_1, \mathbf{X}_2)$ and $\mathbf{A} = (\mathbf{X}_1, \mathbf{X}_2)'(\mathbf{X}_1, \mathbf{X}_2) = \begin{pmatrix} \mathbf{X}_1' \\ \mathbf{X}_2' \end{pmatrix} \begin{pmatrix} \mathbf{X}_1 & \mathbf{X}_2 \end{pmatrix}$. From (28) we see that $\mathbf{A}$ is symmetric, and therefore $\mathbf{A}_{21}' = \mathbf{A}_{12}$ and $\mathbf{B}_{21}' = \mathbf{B}_{12}$. The least squares estimate $\hat{\beta}_1$ is [use (22), (24), (26), (28)]

$$\begin{aligned} \hat{\beta}_1 &= \mathbf{B}_{11}\mathbf{X}_1'\mathbf{y} + \mathbf{B}_{12}\mathbf{X}_2'\mathbf{y} \\ &= \mathbf{B}_{11}\mathbf{X}_1'\mathbf{y} - \mathbf{B}_{11}\mathbf{X}_1'\mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{y} \\ &= (\mathbf{X}_1'\mathbf{X}_1 - \mathbf{X}_1'\mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{X}_1)^{-1}(\mathbf{X}_1' - \mathbf{X}_1'\mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2')\mathbf{y} \\ &= (\mathbf{X}_1'[\mathbf{I} - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2']\mathbf{X}_1)^{-1}\mathbf{X}_1'[\mathbf{I} - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2']\mathbf{y} \\ &= (\mathbf{X}_1'\mathbf{P}\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{P}\mathbf{y}, \end{aligned} \tag{29}$$

where

$$\mathbf{P} = \mathbf{I} - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'$$

is a symmetric projection matrix (it is idempotent – $\mathbf{P}^2 = \mathbf{P}$ – and it is symmetric). Thus, we can rewrite (29) as

$$\hat{\beta}_1 = (\mathbf{X}_1'\mathbf{P}\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{P}\mathbf{y} = ((\mathbf{P}\mathbf{X}_1)'(\mathbf{P}\mathbf{X}_1))^{-1}(\mathbf{P}\mathbf{X}_1)'\mathbf{y}, \tag{30}$$

and also as

$$\hat{\beta}_1 = (\mathbf{X}_1'\mathbf{P}\mathbf{X}_1)^{-1}\mathbf{X}_1'\mathbf{P}\mathbf{P}\mathbf{y} = ((\mathbf{P}\mathbf{X}_1)'(\mathbf{P}\mathbf{X}_1))^{-1}(\mathbf{P}\mathbf{X}_1)'\mathbf{P}\mathbf{y}. \tag{31}$$

Suppose we regress $\mathbf{y}$ on the explanatory variables in $\mathbf{X}_2$. Let $\hat{\gamma}_2$ be the estimate of the corresponding regression coefficients and $\mathbf{v}$ the residual vector from this regression. Then

$$\mathbf{v} = \mathbf{y} - \mathbf{X}_2\hat{\gamma}_2 = \mathbf{y} - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{y} = \mathbf{P}\mathbf{y}.$$

Further, regress each column of $\mathbf{X}_1$ on the explanatory variables in $\mathbf{X}_2$. Place the residuals from these regressions as columns into a matrix, $\mathbf{R}$, so that

$$\mathbf{R} = \mathbf{X}_1 - \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'\mathbf{X}_1 = \mathbf{P}\mathbf{X}_1.$$

Therefore result (31) states that $\hat{\beta}_1$ is obtained by regressing the residuals in $\mathbf{v}$ on the residuals in $\mathbf{R}$, and result (30) states that $\hat{\beta}_1$ is obtained by regressing the response variable itself on the residuals in $\mathbf{R}$.

Note that the matrix $\mathbf{Q} = \mathbf{X}_2(\mathbf{X}_2'\mathbf{X}_2)^{-1}\mathbf{X}_2'$ is also a projection matrix – it is idempotent (and also symmetric). The matrix $\mathbf{P}$ projects onto the space spanned by the residuals, in the sample, and the matrix $\mathbf{Q}$ projects onto the space spanned by the columns of $\mathbf{X}_2$, in the sample. These spaces are orthogonal to each other, which corresponds to

$$\mathbf{P}\mathbf{Q} = \mathbf{Q}\mathbf{P} = \mathbf{0}.$$

(This paragraph is optional) To conclude this discussion, let's consider Problem 2.7 in W. It treats *a partial linear model, a special case of a semiparametric model*. Its projection operations are necessarily conditional expectations, rather than linear projections in the population. Consider the regression function given by the conditional expectation

$$E[y|\mathbf{x}, \mathbf{z}] = g(\mathbf{x}) + \mathbf{z}\beta,$$

where $g(\cdot)$ is a general function of $\mathbf{x}$ and $\mathbf{z}$ is a vector of explanatory variables. The response y is assumed to be

$$y = g(\mathbf{x}) + \mathbf{z}\beta + u, \quad \text{where } E[u|\mathbf{x}, \mathbf{z}] = 0. \quad (32)$$

By iterated expectations we also have $E[u|\mathbf{x}] = 0$. Define the residuals

$$\tilde{y} = y - E[y|\mathbf{x}], \quad \tilde{\mathbf{z}} = \mathbf{z} - E[\mathbf{z}|\mathbf{x}].$$

That is, remove the effects of $\mathbf{x}$ from both y and $\mathbf{z}$. The expectation $E[y|\mathbf{x}]$ is $g(\mathbf{x}) + E[\mathbf{z}|\mathbf{x}]\beta + E[u|\mathbf{x}] = g(\mathbf{x}) + E[\mathbf{z}|\mathbf{x}]\beta$. Thus the y residual is

$$\tilde{y} = g(\mathbf{x}) + \mathbf{z}\beta + u - g(\mathbf{x}) - E[\mathbf{z}|\mathbf{x}]\beta = (\mathbf{z} - E[\mathbf{z}|\mathbf{x}])\beta + u = \tilde{\mathbf{z}}\beta + u. \quad (33)$$

Then

$$E[\tilde{y}|\tilde{\mathbf{z}}] = \tilde{\mathbf{z}}\beta + E[u|\tilde{\mathbf{z}}] = \tilde{\mathbf{z}}\beta.$$

Note that $E[u|\tilde{\mathbf{z}}] = 0$ because $\tilde{\mathbf{z}}$ is a function of $\mathbf{x}$ and $\mathbf{z}$.

This result tells us how we can estimate the parameter β in the model (32). First, estimate $E[y|\mathbf{x}]$ and $E[\mathbf{z}|\mathbf{x}]$ by nonparametric methods. Then form the residuals $\tilde{y}$ and $\tilde{\mathbf{z}}$, and estimate β via (33).

Appendix 1: Proof for the asymptotic normality. Equation (12) is

$$\sqrt{N}(\hat{\beta} - \beta) = \left(\frac{1}{N} \sum_{i=1}^N \mathbf{x}_i' \mathbf{x}_i \right)^{-1} \left(\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{x}_i' u_i \right). \quad (34)$$

In the multivariate case, the Lindeberg-Lévy version of the central limit theorem (Theorem 3.2 in W) states that $1/\sqrt{N}$ times the sum of centered independent and identically distributed random vectors with finite variances and covariances converges in distribution (see Definition 3.6 in W), as N tends to infinity, to $N(\mathbf{0}, \mathbf{B})$, where $\mathbf{B}$ is the covariance matrix of each centred summand. In (34) the factor

$$\frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{x}_i' u_i \quad (35)$$

is a sum of this form. Each summand $\mathbf{x}'_i u_i$ has mean $\mathbf{0}$ by (3) of Assumption 1, and by this result and the assumption of finite second moments, its covariance matrix is

$$\text{Cov}(\mathbf{x}'_i u_i) = E[\mathbf{x}'_i u_i^2 \mathbf{x}] = E[u_i^2 \mathbf{x}'_i \mathbf{x}_i] = \mathbf{B}.$$

Thus, the sequence (35) converges in distribution to $N(\mathbf{0}, \mathbf{B})$. Note that we can also impose Assumption 2 instead of Assumption 1 to obtain this result.

Any sequence which converges in distribution to a nondegenerate limit, is bounded in probability, that is, it is $O_p(1)$ as N tends to infinity, by Lemma 3.5 in W. Thus, the sequence (35) is $O_p(1)$. The other factor on the right-hand side of (34) is the inverse of the matrix

$$\frac{1}{N} \sum_{i=1}^N \mathbf{x}'_i \mathbf{x}_i. \quad (36)$$

By the weak law of large numbers (Theorem 3.1 in W), (36) converges in probability to the expected value of each summand, $E[\mathbf{x}'_i \mathbf{x}_i] = \mathbf{A}$. By Corollary 3.1 in W, the inverse of (36) converges in probability to the inverse of $\mathbf{A}$. That is,

$$\left(\frac{1}{N} \sum_{i=1}^N \mathbf{x}'_i \mathbf{x}_i \right)^{-1} = \mathbf{A}^{-1} + o_p(1). \quad (37)$$

It follows that, as N tends to infinity, the right-hand side of (34) behaves as

$$\begin{aligned} (\mathbf{A}^{-1} + o_p(1)) \frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{x}'_i u_i &= \mathbf{A}^{-1} \frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{x}'_i u_i + o_p(1) \frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{x}'_i u_i \\ &= \mathbf{A}^{-1} \frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{x}'_i u_i + o_p(1) O_p(1) \\ &= \mathbf{A}^{-1} \frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{x}'_i u_i + o_p(1), \end{aligned} \quad (38)$$

where the last step follows from part (4) of Lemma 3.2 in W. By Lemma 3.7 in W (Slutsky's theorem), (38) converges to the same limit distribution as does

$$\mathbf{A}^{-1} \frac{1}{\sqrt{N}} \sum_{i=1}^N \mathbf{x}'_i u_i.$$

The limit distribution is

$$\mathbf{A}^{-1} N(\mathbf{0}, \mathbf{B}) = N(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1}).$$

Appendix 2: The Wald statistic and the hypothesis testing. Let's use the Wald statistic to test a linear hypothesis in OLS estimation.

Consider the model $y = \mathbf{x}\beta + u$ and the null and alternative hypotheses

$$H_0 : \mathbf{R}\beta = \mathbf{r} \quad \text{vs.} \quad H_1 : \mathbf{R}\beta \neq \mathbf{r}, \quad (39)$$

where $\mathbf{R}$ is $q \times K$, $q \leq K$, with row rank q . Moreover, assume all the elements of $\mathbf{x}$ are exogenous.

Assume W's OLS.1 and OLS.2. The OLS estimator $\hat{\beta}$ is asymptotically normal,

$$\sqrt{N}(\hat{\beta} - \beta) \rightarrow_d N(\mathbf{0}, \mathbf{V}),$$

where

$$\mathbf{V} = (E[\mathbf{x}'\mathbf{x}])^{-1} E[u^2 \mathbf{x}'\mathbf{x}] (E[\mathbf{x}'\mathbf{x}])^{-1}.$$

Recall that

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1 \\ \vdots \\ \mathbf{x}_N \end{pmatrix}.$$

A consistent estimator of $\mathbf{V}$ is

$$\hat{\mathbf{V}} = \left(\frac{1}{N} \mathbf{X}'\mathbf{X} \right)^{-1} \frac{1}{N} \sum_{i=1}^N \hat{u}_i^2 \mathbf{x}'_i \mathbf{x}_i \left(\frac{1}{N} \mathbf{X}'\mathbf{X} \right)^{-1},$$

and

$$\hat{\mathbf{V}}/N = (\mathbf{X}'\mathbf{X})^{-1} \sum_{i=1}^N \hat{u}_i^2 \mathbf{x}'_i \mathbf{x}_i (\mathbf{X}'\mathbf{X})^{-1}, \quad (40)$$

where $\hat{u}_i$ is the i th residual from the OLS regression. Then the Wald statistic for testing the null hypothesis is

$$\left(\mathbf{R}\hat{\beta} - \mathbf{r} \right)' \left[\mathbf{R}(\hat{\mathbf{V}}/N)\mathbf{R}' \right]^{-1} \left(\mathbf{R}\hat{\beta} - \mathbf{r} \right), \quad (41)$$

and, under the null hypothesis, this is asymptotically chi-square with q df.

If $q = 1$, then (41) is asymptotically chi-square with 1 df, and we can take the square root and treat the statistic as asymptotically normal. However, it is customary to treat the statistic as having a t -distribution with $N - K$ df, taking into account the finite sample size. If $q > 1$, it is customary to take into account the finite sample size by using an F -statistic in place of the Wald statistic (41). This is accomplished simply by dividing the expression (41) by q , resulting in the F -statistic with q numerator and $N - K$ denominator degrees of freedom. More about this below.

Let's add the assumption of homoscedasticity to the above linear model setting, adding Wooldridge's assumption OLS.3. Then

$$\sqrt{N}(\hat{\beta} - \beta) \rightarrow_d N(\mathbf{0}, \mathbf{V}),$$

where

$$\mathbf{V} = \sigma^2 (E[\mathbf{x}'\mathbf{x}])^{-1}.$$

A consistent estimator of $\mathbf{V}$ is

$$\hat{\mathbf{V}} = \hat{\sigma}^2 \left(\frac{1}{N} \mathbf{X}'\mathbf{X} \right)^{-1},$$

and

$$\hat{\mathbf{V}}/N = \hat{\sigma}^2 (\mathbf{X}'\mathbf{X})^{-1}, \quad (42)$$

where

$$\hat{\sigma}^2 = \frac{1}{N-K} \sum_{i=1}^N \hat{u}_i^2.$$

The Wald statistic for testing the null hypothesis is formed by using (42) in expression (41), resulting in

$$\frac{(\mathbf{R}\hat{\beta} - \mathbf{r})' [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1} (\mathbf{R}\hat{\beta} - \mathbf{r})}{\hat{\sigma}^2}. \quad (43)$$

As above, with the Wald statistic, one treats (43) as chi-square with q df.

Appendix 3: The classical finite sample analysis of OLS estimator with fixed regressor.

Let's return to the model $y = \mathbf{x}\beta + u$. We modify W's assumptions to deal with finite sample size (we assume the classical linear model conditions). First note that we have observations $(y_i, \mathbf{x}_i)$, $i = 1, \dots, N$, and errors u_i , $i = 1, \dots, N$. Write $\mathbf{y} = (y_1, \dots, y_N)'$, $\mathbf{X} = (\mathbf{x}_1', \dots, \mathbf{x}_N')'$ and $\mathbf{u} = (u_1, \dots, u_N)'$.

Assumption A.3.1. $E[u_i|\mathbf{X}] = 0$, $i = 1, \dots, N$, or $E[\mathbf{u}|\mathbf{X}] = \mathbf{0}$.

This assumption guarantees exogeneity for all the explanatory variables.

Assumption A.3.2. The $N \times K$ matrix $\mathbf{X}$ has full column rank, that is, rank K .

This assumption ensures that $\mathbf{X}'\mathbf{X}$ has full rank, equal to K .

Assumption A.3.3. $E[u_i^2|\mathbf{X}] = \sigma^2$, $i = 1, \dots, N$, $E[u_i u_j|\mathbf{X}] = 0$, $i, j = 1, \dots, N$, $i \neq j$.

This assumption is that of homoscedasticity and no correlation between the observations, conditional upon $\mathbf{X}$. We can also state the assumption as

$$\text{cov}(\mathbf{u}|\mathbf{X}) = E[\mathbf{u}\mathbf{u}'|\mathbf{X}] = \sigma^2\mathbf{I}.$$

To consider use of t - and F -distributions, we perform all the following calculations with finite sample size N and *conditional upon* $\mathbf{X}$. Write

$$\mathbf{y} = \mathbf{X}\beta + \mathbf{u}.$$

The OLS estimator of β is

$$\hat{\beta} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y} = \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}.$$

It follows that, conditional upon $\mathbf{X}$, $\hat{\beta}$ is an unbiased estimator of β for every sample size N :

$$\begin{aligned} E[\hat{\beta}|\mathbf{X}] &= E[\beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u} | \mathbf{X}] \\ &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\mathbf{u} | \mathbf{X}] \\ &= \beta + (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{0} \\ &= \beta. \end{aligned}$$

Now, let's find the conditional covariance matrix of $\hat{\beta}$. The calculation is

$$\begin{aligned} E[(\hat{\beta} - \beta)(\hat{\beta} - \beta)' | \mathbf{X}] &= E[(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u}\mathbf{u}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} | \mathbf{X}] \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E[\mathbf{u}\mathbf{u}' | \mathbf{X}]\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\sigma^2\mathbf{I}\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2(\mathbf{X}'\mathbf{X})^{-1}. \end{aligned}$$

The residual vector is

$$\begin{aligned} \hat{u} &= \mathbf{y} - \mathbf{X}\hat{\beta} \\ &= \mathbf{u} - \mathbf{X}(\hat{\beta} - \beta) \\ &= \mathbf{u} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{u} \\ &= [\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}']\mathbf{u} \\ &= \mathbf{P}\mathbf{u}, \end{aligned} \tag{44}$$

and $\mathbf{P} = \mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is symmetric idempotent. The residual vector has conditional mean $\mathbf{0}$, as

$$E[\hat{\mathbf{u}} | \mathbf{X}] = E[(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')\mathbf{u} | \mathbf{X}] = \mathbf{P}E[\mathbf{u} | \mathbf{X}] = \mathbf{P}\mathbf{0} = \mathbf{0}.$$

Next, let's show that $\widehat{\beta}$ and $\widehat{u}$ are uncorrelated, conditional on $\mathbf{X}$. Their conditional covariance is

$$\begin{aligned} E \left[\widehat{\mathbf{u}} \left(\widehat{\beta} - \beta \right)' \mid \mathbf{X} \right] &= E \left[(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') \mathbf{u} \mathbf{u}' \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \mid \mathbf{X} \right] \\ &= (\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') E [\mathbf{u} \mathbf{u}' \mid \mathbf{X}] \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} \\ &= \sigma^2 [\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}] \\ &= \mathbf{0}. \end{aligned}$$

To consider hypothesis testing in this finite sample setting, we make a fourth assumption.

Assumption A.3.4. The distribution of $\mathbf{u}$ conditional on $\mathbf{X}$ is normal, hence $N(\mathbf{0}, \sigma^2 \mathbf{I})$.

Under this assumption we can state that $\widehat{\beta}$ and $\widehat{\mathbf{u}}$ are independent, conditional on $\mathbf{X}$ (for normal variables, zero correlation is equivalent to independence). We will apply this property below.

Let's revisit the linear hypothesis in the present finite sample context,

$$H_0 : \mathbf{R}\beta = \mathbf{r} \quad \text{vs.} \quad H_1 : \mathbf{R}\beta \neq \mathbf{r}. \quad (45)$$

Recall that $\mathbf{R}$ is $q \times K$ and has row rank q . With the addition of Assumption 4, we can state that, conditional on $\mathbf{X}$,

$$\mathbf{R}(\widehat{\beta} - \beta) = \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}' \sim N(\mathbf{0}, \sigma^2 \mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}'), \quad (46)$$

because linear combinations of normal variables are normal. Then it follows from (46) that

$$\frac{(\mathbf{R}(\widehat{\beta} - \beta))' [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1} (\mathbf{R}(\widehat{\beta} - \beta))}{\sigma^2} \quad (47)$$

is, conditional on $\mathbf{X}$, chi-square with q df. To obtain the Wald statistic to test the null hypothesis in (45), we replace σ^2 in (47) by its estimate,

$$\widehat{\sigma}^2 = \frac{1}{N - K} \widehat{\mathbf{u}}' \widehat{\mathbf{u}} = \frac{1}{N - K} \sum_{i=1}^N \widehat{u}_i^2. \quad (48)$$

Next, let's consider this expression.³ The sum of squared residuals divided by σ^2 is

$$\frac{\widehat{\mathbf{u}}' \widehat{\mathbf{u}}}{\sigma^2} = \frac{\mathbf{u}' \mathbf{P}' \mathbf{P} \mathbf{u}}{\sigma^2} = \frac{\mathbf{u}' \mathbf{P} \mathbf{u}}{\sigma^2} = (\mathbf{u}/\sigma)' \mathbf{P} (\mathbf{u}/\sigma). \quad (49)$$

³Note that $\widehat{\sigma}^2$ is an unbiased estimator for σ^2 . First, note that $E[\widehat{\mathbf{u}}' \widehat{\mathbf{u}}] = E[\mathbf{u}' \mathbf{P}' \mathbf{P} \mathbf{u}] = E[\mathbf{u}' \mathbf{P} \mathbf{u}] = \text{tr}(E[\mathbf{u}' \mathbf{P} \mathbf{u}]) = \text{tr}(E[\mathbf{u} \mathbf{u}' \mathbf{P}]) = \text{tr}(\sigma^2 \mathbf{I} \mathbf{P}) = \sigma^2 \text{tr}(\mathbf{I} \mathbf{P}) = \sigma^2 \text{tr}(\mathbf{P}) = \sigma^2(N - K)$. Then, $E[\widehat{\sigma}^2] = E[\frac{1}{N-K} \widehat{\mathbf{u}}' \widehat{\mathbf{u}}] = \sigma^2$.

The vector $\mathbf{u}/\sigma$ is $N(\mathbf{0}, \mathbf{I})$ and $\mathbf{P}$ is symmetric idempotent. Therefore, (49) has a chi-square distribution with degrees of freedom equal to the trace of $\mathbf{P}$,

$$\begin{aligned} \text{tr}(\mathbf{P}) &= \text{tr}(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') \\ &= \text{tr}(\mathbf{I}) - \text{tr}(\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}') \\ &= \text{tr}(\mathbf{I}) - \text{tr}(\mathbf{X}'\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}) \\ &= \text{tr}(\mathbf{I}_N) - \text{tr}(\mathbf{I}_K) \\ &= N - K. \end{aligned}$$

[Note that $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ if the matrices can be multiplied in each order.] From the independence of (48) and (49), we can assert that they are independent chi-squares, conditional on $\mathbf{X}$. And, by (48) and (49), the distribution of

$$\hat{\sigma}^2/\sigma^2, \tag{50}$$

conditional on $\mathbf{X}$, is chi-square with $N - K$ df, divided by $N - K$. Therefore, conditional on $\mathbf{X}$,

$$\frac{\left(\mathbf{R}(\hat{\beta} - \beta)\right)' [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1} \left(\mathbf{R}(\hat{\beta} - \beta)\right) / q}{\sigma^2} \times \left(\frac{\sigma^2}{\hat{\sigma}^2}\right) = \frac{\left(\mathbf{R}(\hat{\beta} - \beta)\right)' [\mathbf{R}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}']^{-1} \left(\mathbf{R}(\hat{\beta} - \beta)\right) / q}{\hat{\sigma}^2} \tag{51}$$

has an F -distribution with q numerator df and $N - K$ denominator df. The only difference between the Wald statistic and this F -statistic is the division by q .⁴

If $q = 1$, we can use the t -test instead of the F -test in this conditional setting. (The square of a t random variable with $N - K$ df is an F random variable with 1 numerator df and $N - K$ denominator df. Recall that the definition of a t random variable with n df is

$$t = \frac{Z}{\sqrt{Y/n}}, \tag{52}$$

where Z is $N(0, 1)$, Y is chi-square with n df, and Z and Y are independent.

Coda. Recall that the mean and variance of a chi-square random variable are equal to the degrees of freedom and twice the degrees of freedom, respectively. Let χ_n^2 denote a chi-square random variable with n df. Conditional on $\mathbf{X}$, the variable (50) has mean

$$E\left(\frac{\chi_{N-K}^2}{N - K}\right) = 1$$

⁴The F -distribution. Assume W_1 is chi-square with n_1 df, W_2 is chi-square with n_2 df, and W_1 and W_2 are independent. Then the distribution of

$$F = \frac{W_1/n_1}{W_2/n_2}$$

is F with n_1 numerator df and n_2 denominator df.

and variance

$$\text{Var} \left(\frac{\chi_{N-K}^2}{N-K} \right) = \frac{2}{N-K}.$$

Therefore, as N tends to infinity, the random variable (50) tends to 1 in probability. It follows that, by Slutsky's theorem, the F random variable (51) converges in distribution to chi-square with q df, divided by q . Similarly, as N tends to infinity, the t random variable (52) converges in distribution to $N(0, 1)$. So, if N is large in hypothesis testing, it is common to use the Wald statistic instead of the corresponding F -statistic, and a normal quantile instead of a t quantile.

Appendix 4. Checking the normality assumption.

Residuals. Assume $(\mathbf{x}_i, y_i)$, $i = 1, \dots, N$, are an independent and identically distributed sample of observations, with $\mathbf{x}_i = (1, x_{i2}, \dots, x_{iK})$. For the model $y = \mathbf{x}\beta + u$, the OLS estimator of β is given at (10),

$$\hat{\beta} = \left(\frac{1}{N} \sum_{i=1}^N \mathbf{x}_i' \mathbf{x}_i \right)^{-1} \left(\frac{1}{N} \sum_{i=1}^N \mathbf{x}_i' y_i \right).$$

The residuals are

$$\hat{u}_i = y_i - \mathbf{x}_i' \hat{\beta}, \quad i = 1, \dots, N.$$

From (10) we can write

$$\mathbf{0} = \sum_{i=1}^N \mathbf{x}_i' y_i - \sum_{i=1}^N \mathbf{x}_i' \mathbf{x}_i \hat{\beta} = \sum_{i=1}^N \mathbf{x}_i' \hat{u}_i.$$

Thus, the residuals obey K linear restrictions,

$$\sum_{i=1}^N x_{ik} \hat{u}_i = 0, \quad k = 1, \dots, K,$$

and one of these restrictions is that the residuals add to zero (this occurs when one of the elements of $\mathbf{x}_i$ is 1, that is, when the model contains an intercept).

Because the residuals are subject to restrictions, they are dependent in finite samples. If N is very large relative to K , the dependence is weak. One can calculate the variances and covariances of the residuals conditional upon the observed values $\mathbf{x}_i$, $i = 1, \dots, N$.

The *standardized residuals* are residuals divided by their standard deviations obtained from this conditional calculation. Use of the standardized residuals is an adjustment often made in finite samples. If one is using asymptotic theory, however, one can consider the residuals to be independent and identically distributed.

The *normal quantile plot*. The purpose of the normal quantile plot is to provide a visual aid to judge whether data are normal. Let's describe construction of the plot. Suppose we have data y_i , $i = 1, \dots, N$. Order the data from smallest to largest, obtaining the ordered values $y^{(i)}$,

$$y^{(1)} \leq y^{(2)} \leq \dots \leq y^{(N)}.$$

Next, we obtain the values we would expect to get in a sample of size N when sampling from a normal distribution. For this purpose we use the standard normal distribution. If Z has a standard normal distribution, the expected values are the normal quantiles z_i , $i = 1, \dots, N$, for which

$$P(Z \leq z_i) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z_i} \exp(-z^2/2) dz = \frac{i - 1/2}{N} = \frac{2i - 1}{2N}, \quad i = 1, \dots, N.$$

Alternatively, we can use probabilities $i/(N + 1)$, $i = 1, \dots, N$, to obtain the normal quantiles z_i , $i = 1, \dots, N$.⁵

Then we plot the points $(z_i, y^{(i)})$, $i = 1, \dots, N$. If the data were drawn from a normal distribution, the points will fall close to a straight line. If we regress $y^{(i)}$ on z_i , the fitted line will have intercept and slope which are estimates of the mean and standard deviation of the normal distribution from which the y_i values were drawn.

Now suppose the $y^{(i)}$ values are the residuals from an OLS regression. The residuals have mean 0 (we assume the model has an intercept), and therefore the fitted line in the plot will have intercept exactly 0 (why is it exactly 0?). If the residual values are well described by a normal distribution, the slope of the fitted line will be an estimate of the standard deviation of the distribution of the residuals (why?). Recall that, if we consider the finite sample properties of the residuals, they are not an independent and identically distributed sample from some distribution, because, conditional upon the observed values $\mathbf{x}_i$, $i = 1, \dots, N$, they do not have identical variances. This is the rationale for using standardized residuals in the normal quantile plot.

More generally, one can construct a quantile-quantile plot (a Q-Q plot) using a distribution other than the normal. The purpose is to provide a visual test that the data are described by the distribution under consideration.

⁵Note that either $\frac{i-1/2}{N}$ or $\frac{i}{(N+1)}$ are a version of the empirical cdf at $y^{(i)}$. Recall that a typical way to define the empirical cdf at $y^{(i)}$ is $\frac{i}{N}$. One problem of using this is that the empirical cdf at $y^{(N)}$ is 1 and therefore z_N becomes infinity for a normal distribution.

Appendix 5: Minimum of the standard error of an estimator. In our discussion of the analysis of the CPS85 data set in one of lab session notes, we calculated the asymptotic standard error of the estimate of the return to education for varying levels of the experience variable. We found that the asymptotic standard error is smallest near the middle of the range of the experience variable. One can give many examples where this result also holds.

Let's consider a simple example and derive an analytical answer. Assume the model is

$$y = E[y|x] + u = \beta_0 + \beta_1 x + \beta_2 x^2 + u.$$

Further, assume $E[u^2|x] = \sigma_u^2$. The estimate of the partial effect is $\hat{\beta}_1 + 2x\hat{\beta}_2$, and its conditional variance is

$$Var(\hat{\beta}_1 + 2x\hat{\beta}_2|x) = Var(\hat{\beta}_1) + 4xCov(\hat{\beta}_1, \hat{\beta}_2) + 4x^2Var(\hat{\beta}_2).$$

This conditional variance is minimized at $x = -Cov(\hat{\beta}_1, \hat{\beta}_2)/[2Var(\hat{\beta}_2)]$. Let's evaluate this minimum value. To do so, we need to determine $Cov(\hat{\beta}_1, \hat{\beta}_2)$ and $Var(\hat{\beta}_2)$. Denote the mean and the variance of x by μ and σ^2 , respectively. We will also need the fourth cumulant of x , which we denote by κ_4 . The asymptotic covariance matrix of $\hat{\beta}$ is estimated by

$$\widehat{Var}(\hat{\beta}) = \hat{\sigma}_u^2 \begin{pmatrix} N & \sum x_i & \sum x_i^2 \\ \sum x_i & \sum x_i^2 & \sum x_i^3 \\ \sum x_i^2 & \sum x_i^3 & \sum x_i^4 \end{pmatrix}^{-1} \approx \frac{\hat{\sigma}_u^2}{N} \begin{pmatrix} 1 & Ex & Ex^2 \\ Ex & Ex^2 & Ex^3 \\ Ex^2 & Ex^3 & Ex^4 \end{pmatrix}^{-1}, \quad (53)$$

where $\hat{\sigma}_u^2$ is the estimate (calculated from the residuals) of the variance of u . The required expectations are

$$\begin{aligned} E[x] &= \mu, \\ E[x^2] &= \sigma^2 + \mu^2, \\ Ex^3 &= E[(x - \mu) + \mu]^3 = E[(x - \mu)^3] + 3\mu\sigma^2 + \mu^3, \\ Ex^4 &= E[(x - \mu + \mu)^4] = E[(x - \mu)^4] + 4\mu E[(x - \mu)^3] + 6\mu^2\sigma^2 + \mu^4 \\ &= \kappa_4 + 3\sigma^4 + 4\mu E[(x - \mu)^3] + 6\mu^2\sigma^2 + \mu^4. \end{aligned}$$

The desired $Cov(\hat{\beta}_1, \hat{\beta}_2)$ and $Var(\hat{\beta}_2)$ are the [2,3] and [3,3] elements of (53). We partition the matrix (53) as

$$\mathbf{B} = \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{pmatrix},$$

where $\mathbf{B}_{22}$ is 2×2 . By the line below (27),

$$\begin{aligned}\mathbf{B}_{22} &= \begin{pmatrix} \text{Var}(\hat{\beta}_1) & \text{Cov}(\hat{\beta}_1, \hat{\beta}_2) \\ \text{Cov}(\hat{\beta}_1, \hat{\beta}_2) & \text{Var}(\hat{\beta}_2) \end{pmatrix} = (\mathbf{A}_{22} - \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12})^{-1} \\ &= \frac{\hat{\sigma}_u^2}{N} \left(\begin{pmatrix} Ex^2 & Ex^3 \\ Ex^3 & Ex^4 \end{pmatrix} - \begin{pmatrix} Ex \\ Ex^2 \end{pmatrix} \begin{pmatrix} Ex & Ex^2 \end{pmatrix} \right)^{-1} \\ &= \frac{\hat{\sigma}_u^2}{N} \begin{pmatrix} \sigma^2 & E[(x - \mu)^3] + 2\mu\sigma^2 \\ E[(x - \mu)^3] + 2\mu\sigma^2 & \kappa_4 + 4\mu E[(x - \mu)^3] + 4\mu^2\sigma^2 + 2\sigma^4 \end{pmatrix}^{-1}.\end{aligned}\tag{54}$$

We find that $\mathbf{B}_{22}$ is

$$C \begin{pmatrix} * & -E[(x - \mu)^3] - 2\mu\sigma^2 \\ -E[(x - \mu)^3] - 2\mu\sigma^2 & \sigma^2 \end{pmatrix},$$

where C is $\hat{\sigma}_u^2/N$ divided by the determinant of the 2×2 matrix to be inverted in (54). Therefore the minimum of the asymptotic standard error occurs at

$$\mu + \frac{E[(x - \mu)^3]}{2\sigma^2}.\tag{55}$$

It is interesting that this result does not depend on the fourth cumulant of x . If the distribution of x is close to symmetric, the minimum will be close to μ .

The result we have obtained is somewhat limited. Let's try to extend it. Suppose the model we fit is

$$y = E[y|x, \mathbf{z}] + u = \beta_0 + \beta_1 x + \beta_2 x^2 + \mathbf{z}\gamma + u,\tag{56}$$

where $\mathbf{z}$ is a vector of covariates. Rewrite (56), by adjusting for the mean of $\mathbf{z}$, denoted by $\mu_{\mathbf{z}}$, as

$$y = (\beta_0 + \mu_{\mathbf{z}}\gamma) + \beta_1 x + \beta_2 x^2 + (\mathbf{z} - \mu_{\mathbf{z}})\gamma + u.$$

We do estimation in this model with $\mu_{\mathbf{z}}$ replaced by the vector of sample averages. The asymptotic covariance matrix of the parameter vector $(\beta', \gamma')'$ is approximately equal to [see(53) for comparison]

$$\frac{\hat{\sigma}_u^2}{N} \begin{pmatrix} 1 & Ex & Ex^2 & \mathbf{0}' \\ Ex & Ex^2 & Ex^3 & E[x(\mathbf{z} - \mu_{\mathbf{z}})] \\ Ex^2 & Ex^3 & Ex^4 & E[x^2(\mathbf{z} - \mu_{\mathbf{z}})] \\ \mathbf{0} & E[x(\mathbf{z} - \mu_{\mathbf{z}})]' & E[x^2(\mathbf{z} - \mu_{\mathbf{z}})]' & \text{Cov}(\mathbf{z}') \end{pmatrix}^{-1}.\tag{57}$$

If we assume x and x^2 are uncorrelated with $\mathbf{z}$, (57) reduces to $(\beta', \gamma')'$ is approximately equal to [see(53) for comparison]

$$\frac{\hat{\sigma}_u^2}{N} \begin{pmatrix} 1 & Ex & Ex^2 & \mathbf{0}' \\ Ex & Ex^2 & Ex^3 & \mathbf{0}' \\ Ex^2 & Ex^3 & Ex^4 & \mathbf{0}' \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \text{Cov}(\mathbf{z}') \end{pmatrix}^{-1},$$

and thus the minimum of the asymptotic standard error of the estimate of the partial effect of x occurs at (55). If the correlations between x , x^2 and $\mathbf{z}$ are small, then the minimum will be close to (55).